Document Name: Master Batch Record: Production Bioreactor and Triple Transfection				Page 1
Number: MBR-USP-201	Version: 1.0	Effective Date: 01APR2026	Status: Effective	Part No: DS-OBX-201

Product and Lot Information

Product		Drug substance lot	
Process step		Manufacturing date	
Cell line		Working cell bank lot	
Production bioreactor ID		Working volume (mL)	
Transfection reagent		Target harvest	

Referenced SOPs

SOP Number	Title
SOP-3010	Working Cell Bank Thaw and Seed Train Expansion
SOP-3040	N-1 Seed Bioreactor Operation
SOP-3110	Triple Transfection and Production Culture
SOP-3120	Production Bioreactor In-Process Monitoring
SOP-3150	Harvest and Lysis
SOP-1203	Buffer and Solution Preparation
SOP-0905	Bioburden and Endotoxin Sampling

1. EQUIPMENT / INSTRUMENTS

Description	EIN	Calibration Due	Performed By / Date
Production bioreactor skid			
Dissolved oxygen probe			
pH probe			
Viable cell counter			
Analytical balance			
Biosafety cabinet			

2. PLASMID AND REAGENT PREPARATION

Component	Identity / Composition	Lot Number	Prepared By / Date	Verified By / Date

Rep/Cap plasmid	pOBX-RepCap (AAV9)			
Helper plasmid	pHelper (Ad E2A, E4, VA RNA)			
Transgene plasmid	pOBX-ITR-transgene			
Transfection reagent	Linear PEI, 1 mg/mL			
Production medium	Chemically defined, serum-free			

3. SEED AND INOCULATION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
3.1	Record viable cell density at inoculation.	Viable cell density (e6 cells/mL). Acceptance: 0.5 to 1.5.			
3.2	Record viability at inoculation.	Viability (percent). Acceptance: not less than 90.			
3.3	Confirm working volume.	Working volume (mL). Acceptance: 10,000 plus or minus 500.			

4. TRANSFECTION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
4.1	Record viable cell density at transfection.	Viable cell density (e6 cells/mL). Acceptance: 1.0 to 2.0.			
4.2	Record culture volume at transfection.	Culture volume (mL). Acceptance: record.			
4.3	Calculate total plasmid DNA for the transfection.	Total DNA (mg). Acceptance: 1.0 microgram/mL per SOP-3110. Formula: Total DNA (mg) = DNA per mL (microgram/mL) x Culture volume (mL) / 1000	DNA per mL (microgram/mL): Culture volume (mL): Result (mg):		

4.4	Calculate PEI mass from the DNA to PEI ratio.	PEI mass (mg). Acceptance: DNA to PEI 1:2 by weight per SOP-3110. Formula: PEI (mg) = Total DNA (mg) x 2	Total DNA (mg): DNA:PEI ratio: Result (mg):		
4.5	Split plasmid DNA by molar ratio.	Rep/Cap : Helper : Transgene = 1:1:1 (mg each).			
4.6	Record DNA-PEI complexation time before addition.	Complexation time (min). Acceptance: 15 to 30 min.			
4.7	Confirm addition of transfection complex to the reactor.	Complex added to reactor?	☐ Yes ☐ No		

5. PRODUCTION CULTURE

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
5.1	Record temperature setpoint.	Temperature (C). Acceptance: 37.0 plus or minus 0.5.			
5.2	Record dissolved oxygen setpoint.	Dissolved oxygen (percent). Acceptance: 40 plus or minus 10.			
5.3	Record hour 24 viability.	Viability (percent). Acceptance: record.			
5.4	Record hour 24 culture pH.	pH. Acceptance: 7.0 to 7.2.			
5.5	Record hour 48 viability.	Viability (percent). Acceptance: record.			
5.6	Record hour 48 culture pH.	pH. Acceptance: 7.0 to 7.2.			
5.7	Record hour 72 viability.	Viability (percent). Acceptance: record.			

6. HARVEST

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
6.1	Record harvest time post-transfection.	Harvest time (h). Acceptance: 48 to 96, target 72.			
6.2	Record harvest viability.	Viability (percent). Acceptance: not less than 70.			
6.3	Record harvest working volume.	Volume (mL). Acceptance: record.			

7. IN-PROCESS SAMPLING

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
7.1	Record in-process vector genome titer (qPCR).	VG titer (e10 vg/mL). Acceptance: record.			
7.2	Confirm bioburden sample collected.	Bioburden sample collected?	☐ Yes ☐ No		
7.3	Confirm mycoplasma sample collected.	Mycoplasma sample collected?	☐ Yes ☐ No		

8. PROCESS COMMENTS

Page / Step	Comment	Performed By / Date	Verified By / Date

BATCH RECORD APPROVALS

Manufacturing Review	Quality Assurance Review

Manufacturing and QA review signatures are applied at batch disposition.